UNICITY

SECURING THE AGENTIC ECONOMY

A STRATEGIC SEED PROPOSAL FOR GREG KIDD · HARD YAKA

The autonomous machine economy is bottlenecked by fragmented authorization and slow settlement infrastructure. Unicity binds the proof of **identity** and **permission** directly to the payment asset itself.

THE ACTOR HAS CHANGED: THE RISE OF MACHINE FINANCE

Autonomous agents now initiate high-frequency, micro-value transactions at scale.

57.5%

OF WEB REQUESTS ARE AUTOMATED, NOT HUMAN — CLOUDFLARE, 2026

Traditional infrastructure cannot settle that volume securely, and the failure point is **trust**. Standard payment networks cannot natively verify an agent's authorization, enforce a spending limit, or prove a recipient's identity without slow central intermediaries. Unicity makes the cryptographic **proof of authority travel with the money**.

THE STRUCTURAL BOTTLENECK: THE SHARED LEDGER

Global consensus architectures cannot physically process machine-speed commerce.

Traditional blockchains require every node to broadcast, order, validate, and record every transaction. This shared-ledger design imposes a permanent ceiling on throughput and privacy. True **peer-to-peer settlement** requires eliminating the shared ledger entirely.

BROADCAST

Every node hears every transaction.

ORDER

Global agreement on the sequence of all of them.

VALIDATE

Every node re-runs every one.

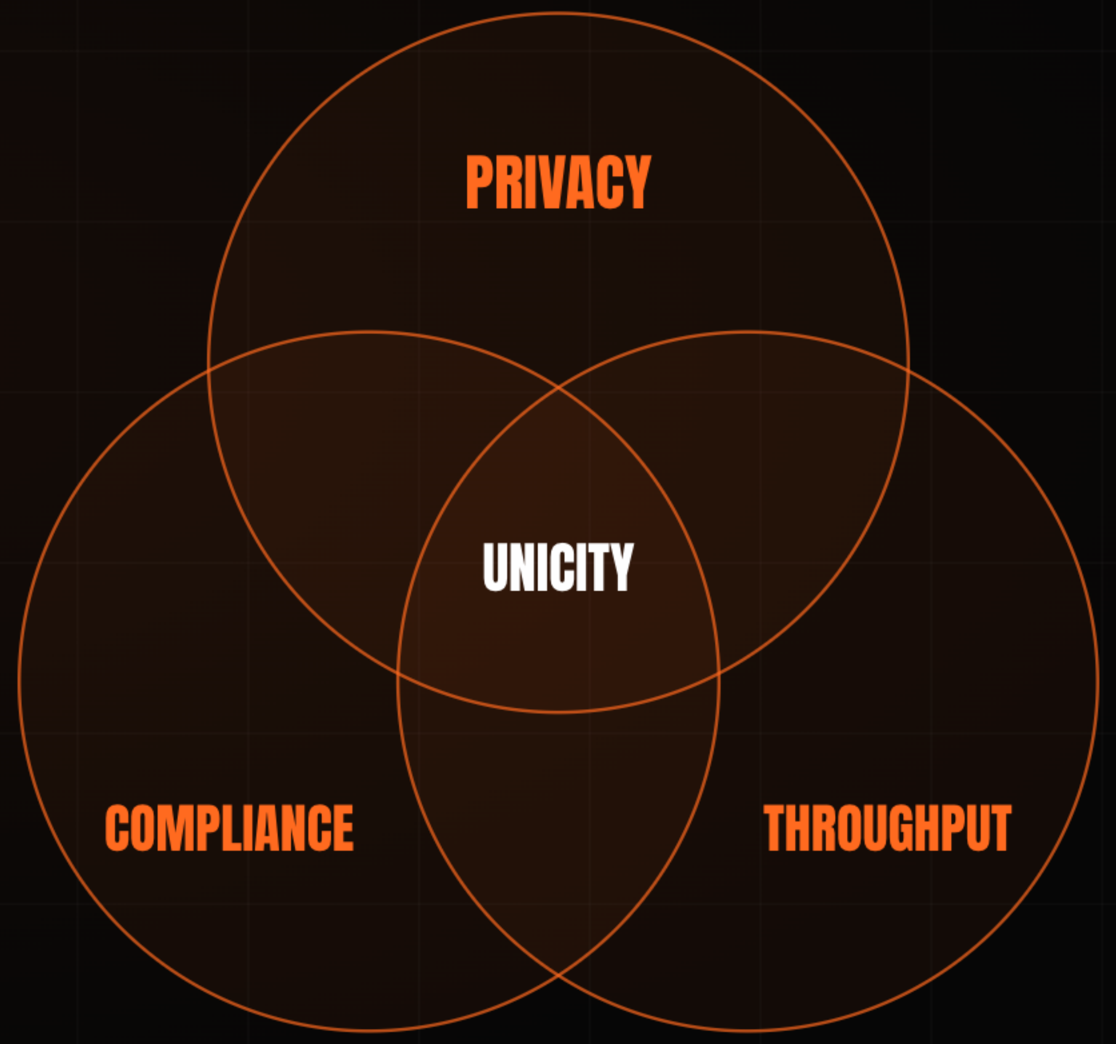
RECORD

Every node stores the whole state, forever.

THE STABLECOIN TRILEMMA: PICK TWO

Every digital-dollar design has been forced to sacrifice one of compliance, privacy, or throughput.

A public ledger delivers compliance and throughput, but it exposes every balance and counterparty. Add zero-knowledge proofs and privacy returns, but throughput collapses under the proving cost. Greg Kidd's payments portfolio lives inside this trade-off. Unicity is the first protocol to hold **all three at once**, because it never places the transaction on a shared ledger to begin with.



THE FINAL INFRASTRUCTURE GAP: EMBEDDED IDENTITY

The market solved disintermediation and value transfer. It has not solved decentralized counterparty verification.

01	**DISINTERMEDIATION** Open infrastructure bypassed the legacy financial middleman.	SOLVED
02	**DIGITAL VALUE** Stablecoins now settle tens of trillions of dollars a year.	SOLVED
03	**CRYPTOGRAPHIC IDENTITY** No blockchain can natively verify a recipient's legal or operational standing before a transaction executes.	UNSOLVED

Unicity was engineered to close the third gap.

FROM SIDE-CAR TO PROTOCOL: EVOLVING DIGITAL IDENTITY

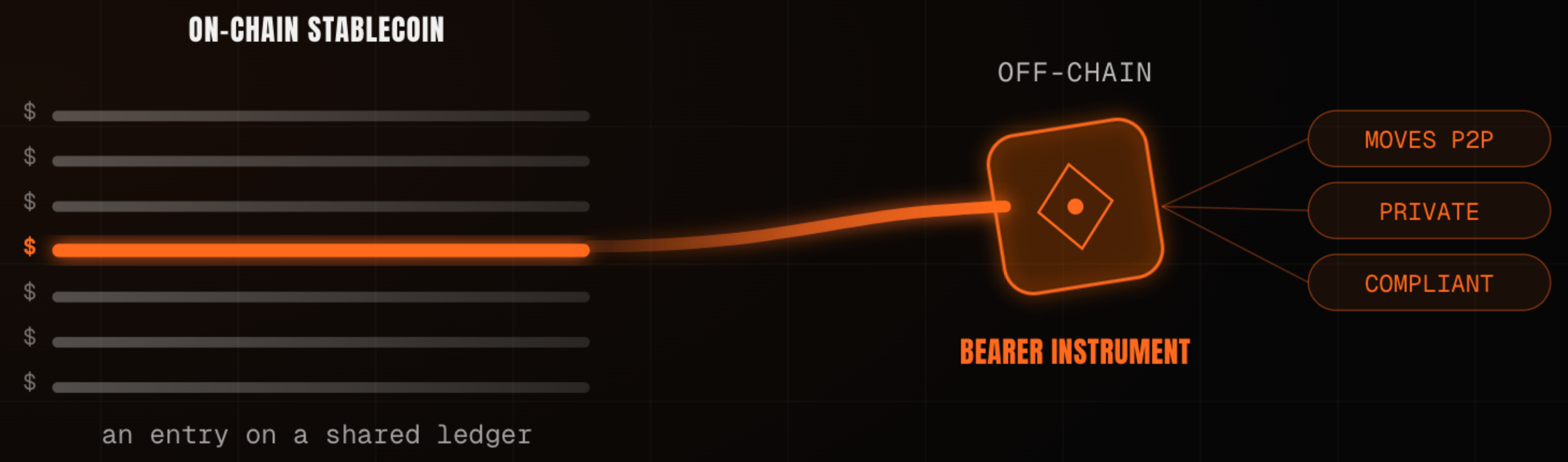
Verification must be built into the asset, not layered on as an optional application check.

Early self-sovereign identity stalled because the credential sat parallel to the transaction, and a check that can be skipped gets skipped. USBC drew the right conclusion: identity and compliance must reside **inside the dollar itself**. Unicity supplies the underlying protocol that makes embedded identity strictly enforceable across any jurisdiction.

FROM LEDGER ENTRIES TO BEARER INSTRUMENTS

Off-chain settlement restores the properties of physical cash to digital stablecoins.

A standard on-chain stablecoin is only an entry on a shared ledger. Unicity converts it into a **self-contained, self-proving bearer instrument**. The value and the proof of validity live within the object, so the asset moves **peer-to-peer** across any transport layer (NOSTR, HTTP, QR) with no centralized ledger lookup.



THE UNIQUENESS ORACLE: UNBUNDLING CONSENSUS

Each generation of consensus removed work from the network. Unicity keeps only the irreducible function.

Unicity introduces the **Uniqueness Oracle**, which attests to one thing: has this token been spent? Because the Oracle never re-executes transactions or reads payloads, throughput scales horizontally — **by design, 30,000 transactions per second per shard** at sub-microcent cost.

CONSENSUS & EMISSION
decentralized base security

↓ certified state root

UNIQUENESS ORACLE
stores only proofs that token states are unique

↓ inclusion proofs

EXECUTION LAYER
agents and users transact client-side, off-chain

2009 · BITCOIN

CORRECTNESS + ORDERING

Every node certifies every transaction and agrees the global order.

2023 · SUI · FASTPAY

CORRECTNESS ONLY

Global ordering dropped for assets that share no state.

2026 · UNICITY

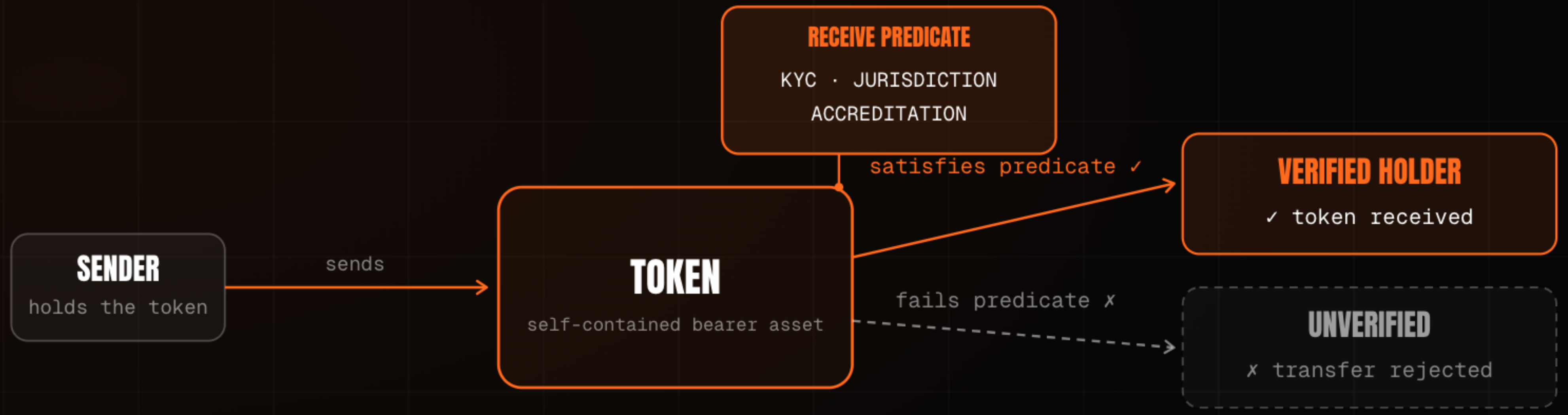
UNIQUENESS ONLY

The network attests one thing. Correctness moves to the edge.

PROTOCOL-ENFORCED COMPLIANCE: THE RECEIVE PREDICATE

Cryptographically gating asset transfers without sacrificing peer-to-peer privacy.

Standard compliance puts a central facilitator on a public ledger, which breaks enterprise privacy. Unicity uses the **Receive Predicate** instead. KYC, jurisdiction, and accreditation are programmed by the issuer directly into the bearer asset and enforced before receipt is allowed. If the recipient cannot satisfy the predicate locally, the transfer cannot execute. **The asset enforces its own compliance.**



PRIVACY BY CONSTRUCTION: PROVEN AGAINST EVERY OBSERVER

Confidentiality is a founding property of the protocol, not a check added at the application layer.

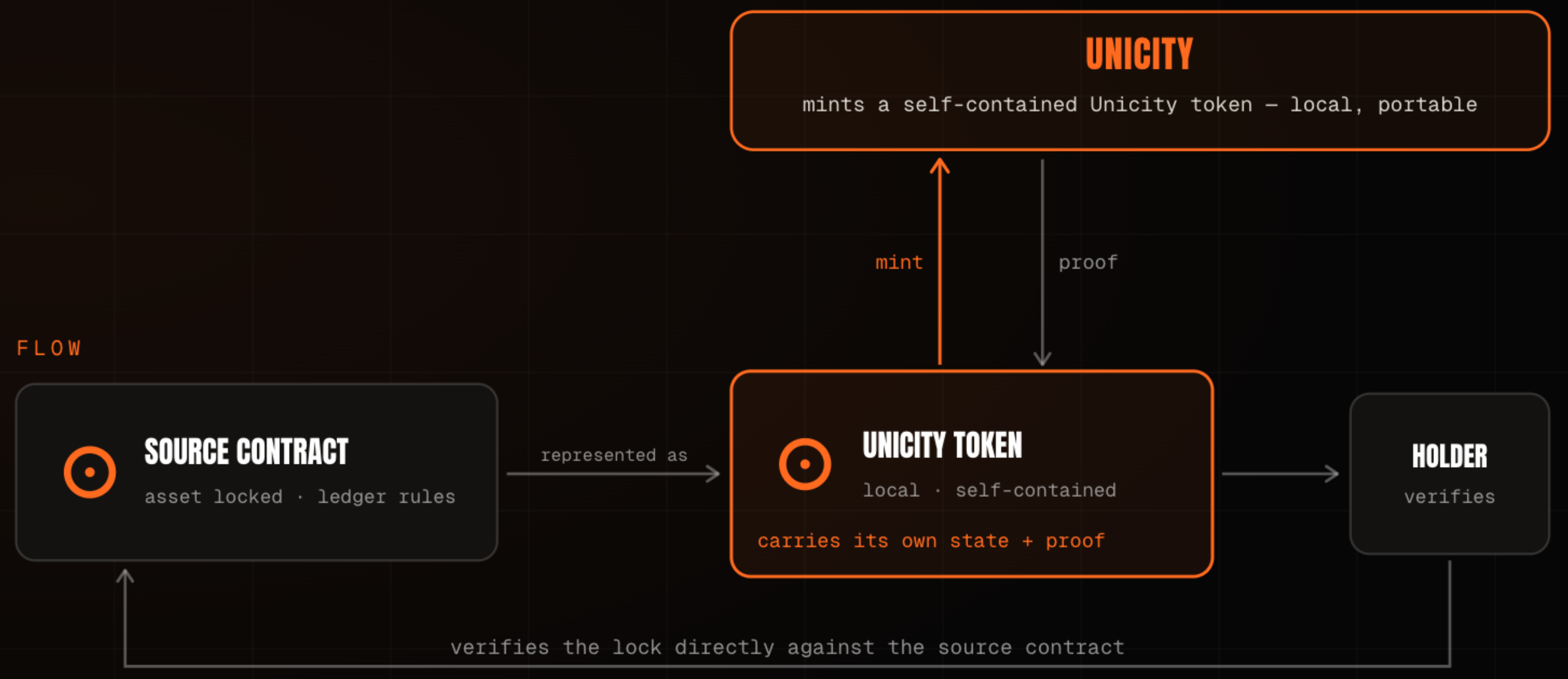
The privacy holds against all three observers in any transaction, and the mathematical papers prove it — which is what makes Unicity viable for the institutional and machine-to-machine flows that public ledgers expose.

THE NETWORK	Sees only opaque commitments. It cannot read amounts, parties, or balances, nor link one transfer to the next.	PROVEN
THE SENDER	Cannot watch where or when the recipient later spends what was sent.	PROVEN
ANYONE WITH THE ADDRESS	Sees one published key; every sender derives a fresh, indistinguishable transaction key.	PROVEN

NO BRIDGE, NOTHING TO HACK

Cross-chain bridges are the largest loss category in crypto because they custody the asset and forward a forgeable message.

A bridge concentrates risk by holding value and trusting a relayed instruction. Unicity holds nothing. A locked source-chain asset becomes a self-contained bearer token, and the recipient verifies the lock directly against the source contract. No cross-chain message exists to forge, no pooled liquidity to drain, and no custodian to compromise. The asset **proves its own provenance**.



THE ATOMIC SWAP: SETTLEMENT WITHOUT AN INTERMEDIARY

The hardest problem in cash-like money is the trade, where two parties exchange value with no one holding both legs.

The conventional fix is a timed lock, but the clock becomes the attack surface, because one side can stall and walk away. Unicity removes the clock. Both parties commit independently, and the swap either forms for both at once or not at all. Settlement runs off-chain at machine speed with **no mempool and no MEV** — the precondition for autonomous agents trading directly with one another.

HASHED TIMELOCK (HTLC)

1 · A locks, reveals hash y

2 · B locks against same y

3 · A claims, revealing preimage x

4 · B must claim before timeout

THE TIMING TRAP

B late on step 4 — a dropped connection — and A keeps both.

UNICITY PREDICATE SWAP

A COMMITS
locks their token

B COMMITS
locks their token

UNICITY SERVICE
the shared reference

AUTOMATIC FAIRNESS

- ✓ both locked → swap completes
- ✓ either walked away → both refunded
- ✓ no deadlines, no chasing, done

X402, REBUILT: TWELVE STEPS TO FIVE

Removing the facilitator and the shared ledger cuts the agent-payment handshake by more than half.

The x402 standard lets an agent pay for a resource over HTTP, but settlement still routes through a facilitator and a public chain. The bearer token carries its own authorization and proof, so the payment clears directly between client and server — the kind of friction reduction a payments operator like Square measures in basis points.

TRADITIONAL X402 — 12 STEPS

1	Client requests the resource
2	Server returns 402 Payment Required
3	Client signs and submits payment
4	~~Server forwards to facilitator~~
5	~~Facilitator verifies on-chain balance~~
6	~~Settle on the shared ledger~~
7	~~Await block confirmation~~
8	~~Facilitator confirms to server~~
…	then deliver the resource

UNICITY X402 — 5 STEPS

1	Client requests the resource
2	Server returns 402 Payment Required
3	Client pays with a bearer token (auth + proof inside)
4	Server verifies the token locally
5	Server delivers the resource

7 STEPS ELIMINATED

No facilitator. No shared ledger. No block wait.

THE MACHINE MARKET: THE VENUE DISAPPEARS

When two agents settle directly, the exchange in the middle has nothing left to do.

Centralized exchanges deliver speed but custody the assets. Decentralized exchanges preserve self-custody but expose every order publicly and were never built for a machine counterparty. Unicity combines exchange-grade speed, self-custody, privacy, and protocol-enforced compliance with no venue at all.

	BINANCE	UNISWAP	HYPERLIQUID	UNICITY
SPEED	sub-second	block time	fast	sub-second
SELF-CUSTODY	custodial	you hold keys	you hold keys	you hold keys
PRIVACY	operator sees all	fully public	public positions	unlinkable
COMPLIANCE	off-chain KYC	none	none	in the asset

WHAT GETS BUILT: THE AGENTIC CORPORATION

Agents become the new smart contracts, executing verifiable logic directly on bearer assets.

Unicity built a decentralized autonomous corporation for BlackRock: a weather-based parametric insurer whose capital, underwriting, and reinsurance cession all run as agents transacting in Unicity tokens. The logic is verifiable code, the settlement is peer-to-peer, and the compliance travels inside each asset. This is the natural application of **identity-bound, machine-speed settlement**.

THE INCUMBENTS VALIDATE THE DIAGNOSIS

The largest payments and crypto players are spending billions to work around the ledger architecture they built.

Google's AP2 authorizes the agent and Coinbase's x402 settles the payment, while the major chains optimize the ledger underneath. Each layer holds one fragment of the proof, so the mandate and the audit trail scatter across separate systems. **Unicity keeps the entire proof inside the asset.**

PLAYER	THEIR MOVE	APPROACH
CIRCLE	Pivoted to infrastructure — the Arc L1, a $222M presale, bought Malachite.	optimise the ledger
SOLANA	Alpenglow cut finality to ~150ms; its co-founder concedes a physical ceiling.	optimise the ledger
AP2 · X402	Authorize in one layer, settle in another — the proof splits across the stack.	split the proof
UNICITY	Keeps authorization, settlement, and the audit trail inside the asset.	keep it whole

THE TEAM: SOVEREIGN-GRADE VERIFICATION, BUILT BEFORE

The architects shipped nation-scale cryptographic infrastructure long before agentic finance had a name.

THE TEAM

Built Guardtime's **KSI** — the verification layer deployed with the Estonian Government, Lockheed Martin, Boeing and NATO. In production since 2012, eIDAS-grade. That lineage sustained **300,000 tx/sec** in Eesti Pank's 2021 test (heritage, not a live Unicity figure).

THE PROOF

Three mathematical papers prove the **privacy** and **no-double-spend** guarantees. Drop them into any model and they check. All public on github.com/unicitynetwork. The proof is verifiable today, not asserted.

THE ALIGNMENT

Identity, speed, and peer-to-peer settlement, bound into one instrument — the fair-access infrastructure Hard Yaka has backed across a generation of payments companies.

THE ASK: $5M TO SETTLE THE FIRST COMPLIANT DOLLAR

A seed round to ship a USBC-style dollar that settles the instant the credential checks out.

$5M Seed · $25M cap / $50M token FDV · SAFE + token warrant

The first build is a compliant dollar that settles peer-to-peer the moment the Receive Predicate is satisfied, with the partner supplying the regulatory layer. This binds **identity, speed, and settlement** into a single instrument — precisely the fair-access infrastructure Hard Yaka has backed across a generation of payments companies.

One charter settles at home. Unicity carries the same rule between any two parties that share nothing else.